

J. Baeden Vester

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EDUCATION

North Carolina State University

Master of Computer Science

Raleigh, NC

Aug. 2025 - Present

- **Coursework:** Software Engineering, Robotics, Operating Systems, Cybersecurity

University of North Carolina at Chapel Hill

Bachelor of Science in Computer Science, Minor in Chinese

Chapel Hill, NC

Aug. 2022 - May 2025

- **Coursework:** Algorithms, Data Structures, Databases, Models of Languages and Computation, Computer Organization, Computer Vision, Numerical Analysis

EXPERIENCE

Product Developer

Sep. 2023 - Present

CircuitSays

Raleigh, NC

- Owned the full development lifecycle – design, coding, testing, and debugging – of a **React-based** companion web app for physical electronics kits, driving **10 new client schools** into the grant process and **over \$20,000** in standalone grants.
- Implemented a testing suite with **80% code coverage**, and partnered with teachers and prospective client schools to gather feedback that strengthened product-market fit.
- Accelerated feature delivery for client demos by collaborating directly with the hardware team, aligning software UI/UX with physical kit design across the product lifecycle.

PROJECTS

CalculateTripCost - Driving Expense Tracker | *React, FastAPI, PostgreSQL, Tailwind CSS, Chart.js*

- Engineered a full-stack web application end-to-end with a **React frontend** and **Python (FastAPI) backend** to track and analyze driving expenses with real-time cost calculations.
- Designed a **PostgreSQL** schema with optimized queries for statistical analysis, integrated the **U.S. EIA API** for live gas prices, and visualized trends with **Chart.js**.
- Built a responsive **Tailwind CSS** UI with seamless mobile/desktop compatibility.

Connect Four RL | *Python, PyTorch, Reinforcement Learning*

- Built a Connect Four game with an AI opponent powered by a neural net trained via self-play against an **Alpha/Beta Pruned Minimax** tree.
- Designed the training pipeline and evaluation loop in **Python/PyTorch** to iteratively strengthen the neural net against the minimax opponent.
- Increased the agent's win rate over successive training runs by tuning reward shaping and search depth, validated through repeated head-to-head evaluation matches.

Multi-Drone Control | *Python, Isaac Sim, Isaac Lab, PX4, MAVLink, QGroundControl, Pegasus*

- Built a modular simulation framework in **Python** and **NVIDIA Isaac Sim** for simultaneous programmatic control of multiple drones using both high-level flight commands and low-level motor control.
- Implemented dual control architectures supporting **MAVLink** commands via PX4/QGroundControl alongside a **custom PID controller** for direct motor-level control.
- Reduced onboarding time for new contributors by authoring modular documentation covering PX4 controller creation, direct motor control, and RL training workflows.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL/SQLite, C/C++, Rust, Bash, HTML/CSS

Frameworks/Libraries: React, FastAPI, Node.js, Vite, PostgreSQL, NumPy, PyTorch, Matplotlib, OpenCV, Langchain, OpenGL, Tkinter, Slint

Developer Tools: Linux, Git, GitHub Actions, Docker, Make, Neovim, VS Code, ChatGPT, Claude, Ollama